SVM · MARGIN & OBJECTIVE

SVM: Margin and Objective

Distance, margin width, and four forms of the objective

From the point-to-line formula to the dual

Where 2 / ||w|| comes from, why maximizing it means minimizing ½||w||²,
and how soft margin, hinge loss, and the dual extend the problem.

Three ideas carry the whole derivation

Each later slide expands one of these statements

1

SVM distance is the ordinary point-to-line distance

Substitute the point into the boundary expression, take the absolute value, divide by ||w||.

2

The full margin equals 2 / ||w||

H_+ and H_- sit at +1 and −1; each is 1 / ||w|| from the decision boundary.

3

The objective has four equivalent or extended forms

Hard margin, soft margin, hinge loss, and the dual each serve a different purpose.

Margin width = 2 / ||w|| → hard-margin SVM minimizes ½||w||²

Four terms are needed before any formula

Definitions used throughout the deck, stated for a linear SVM

Hyperplane

$w^T x + b = 0$

A flat boundary that splits the feature space into two sides. A line in 2-D, a plane in 3-D.

Normal vector w

perpendicular to the boundary

Sets the boundary's orientation. Its length $||w||$ scales every value of $w^T x + b$.

Margin

empty band around the boundary

The distance between the two parallel boundaries H_+ and H_- that bracket the data.

Support vectors

points on H_+ or H_-

The training points that touch the margin and fix where it lies.

Distance is the line expression evaluated at the point

Point P = (x_0, y_0) and line ax + by + c = 0, with a and b not both zero

$$d = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$$

Numerator

- Substitute the point into the whole left side of the line equation
- Take the absolute value: distance is never negative

Denominator

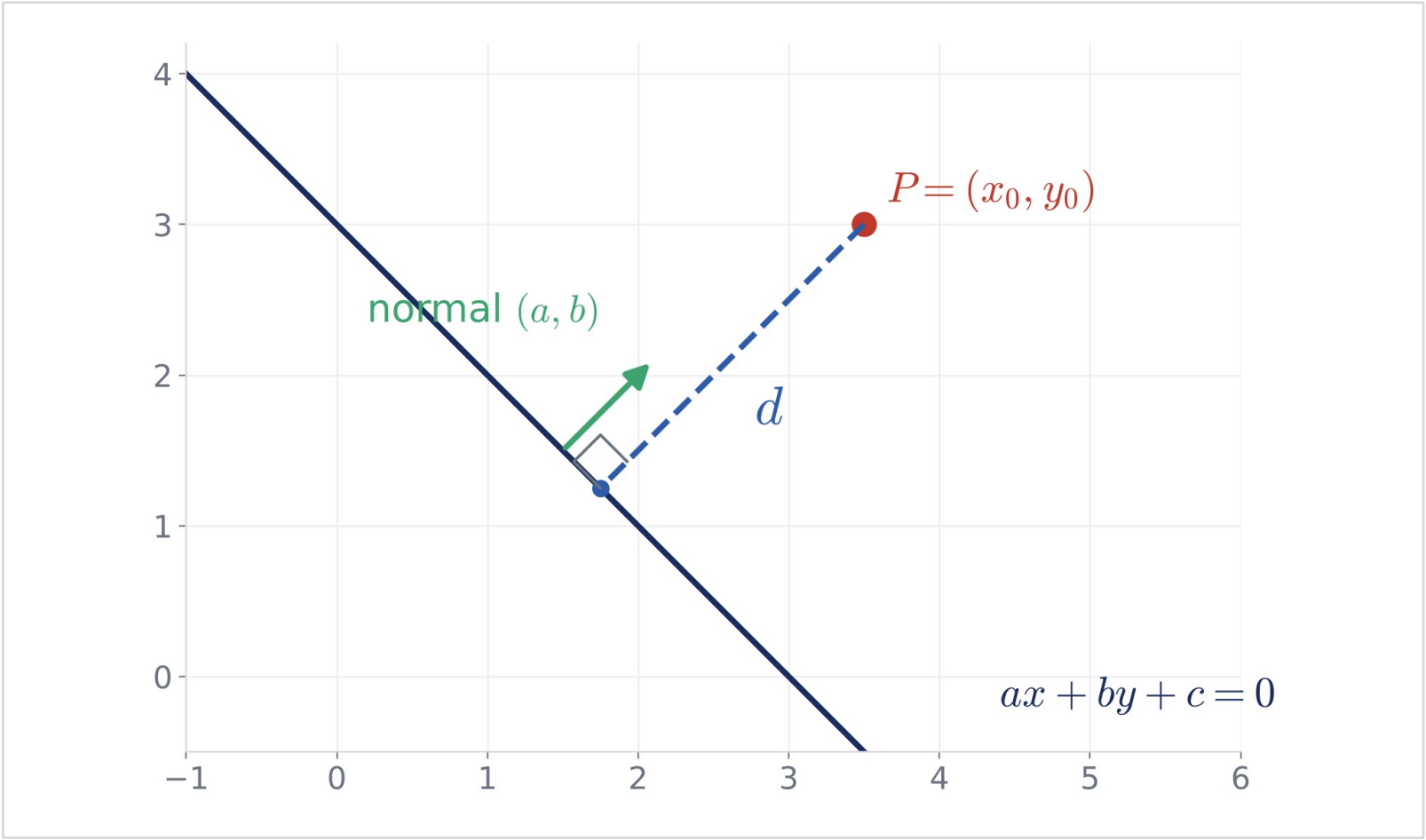
- √(a² + b²) is the length of the normal (a, b)
- It removes the effect of rescaling the coefficients

The shortest path runs along the normal

The dashed segment meets the line at a right angle

How to read

- Red point: P
- Green arrow: normal (a, b)
- Dashed segment: d
- Right angle at the foot



The SVM distance is the same formula in vector form

Decision boundary $w^T x + b = 0$; w is the normal vector and b the intercept

$$d(\mathbf{x},\ \text{boundary}) = \frac{|\mathbf{w}^\top \mathbf{x} + b|}{\|\mathbf{w}\|}$$

$$\mathbf{w} = (w_1, w_2), \qquad \|\mathbf{w}\| = \sqrt{w_1^2 + w_2^2}$$

Mapping

- (a, b) becomes w, c becomes b

Scope

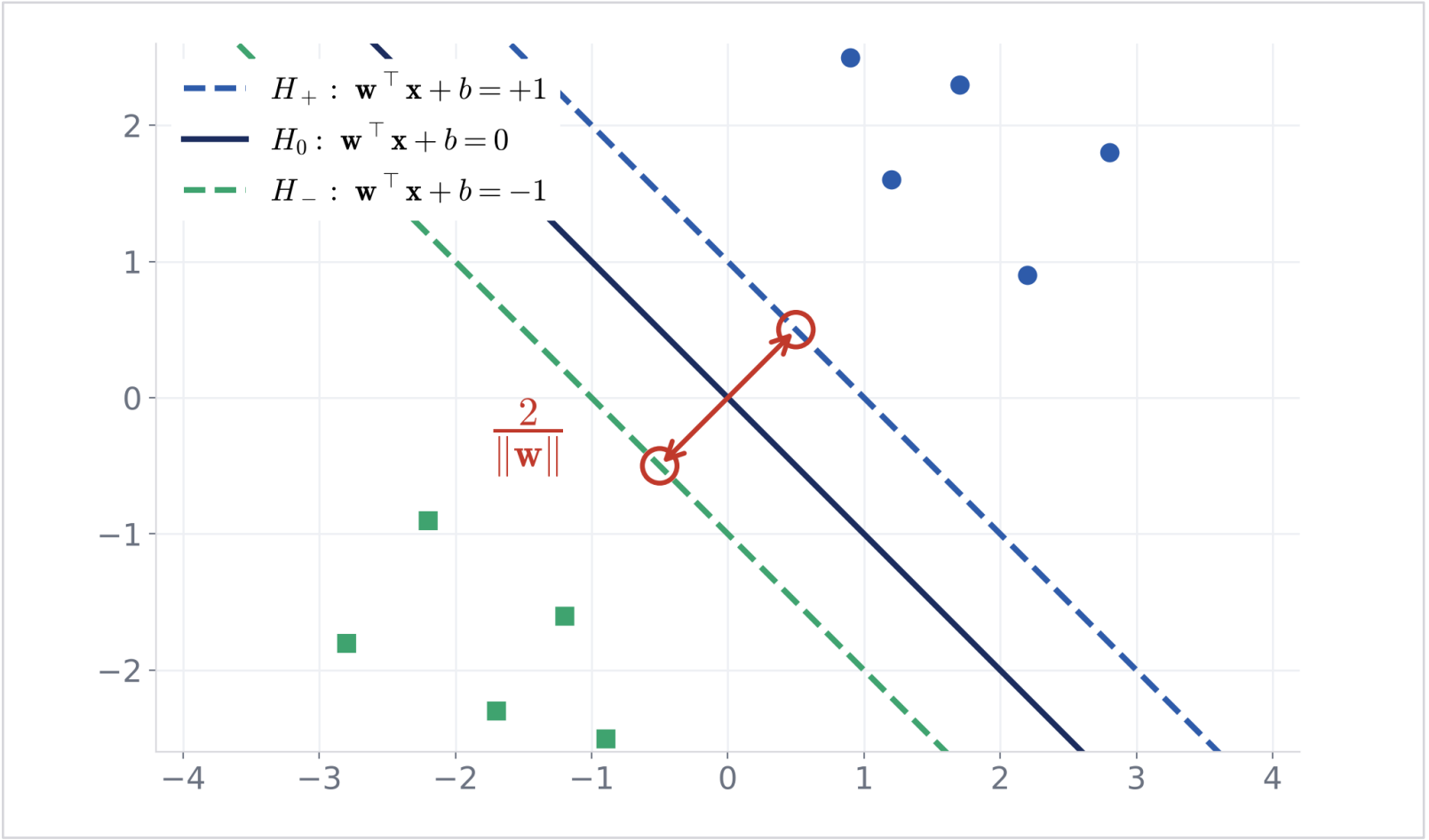
- Holds for hyperplanes in any dimension

Margin boundaries sit on both sides of H_0

Standard SVM scaling puts them at $w^T x + b = +1$ and -1

How to read

- Solid line: H_0
- Dashed lines: H_+, H_-
- Circled points: support vectors
- Red arrow: margin



The full margin equals 2 / ||w||

Each margin boundary is 1 / ||w|| from the central decision boundary

$$H_+ : \ \mathbf{w}^\top \mathbf{x} + b = +1$$
$$H_0 : \ \mathbf{w}^\top \mathbf{x} + b = 0$$
$$H_- : \ \mathbf{w}^\top \mathbf{x} + b = -1$$

$$\underbrace{\frac{1}{\|\mathbf{w}\|}}_{H_+ \to H_0} + \underbrace{\frac{1}{\|\mathbf{w}\|}}_{H_0 \to H_-} = \frac{2}{\|\mathbf{w}\|}$$

Next slide: the same 2 / ||w|| obtained in a single distance calculation

One distance calculation gives the whole margin

Take a point x_+ on H_+ and measure its distance to H_-

$$(1) \quad \mathbf{w}^\top \mathbf{x}_+ + b = 1$$

$$(2) \quad \mathbf{w}^\top \mathbf{x} + b = -1 \implies \mathbf{w}^\top \mathbf{x} + b + 1 = 0$$

$$(3) \quad d = \frac{|\mathbf{w}^\top \mathbf{x}_+ + b + 1|}{\|\mathbf{w}\|} = \frac{|1 + 1|}{\|\mathbf{w}\|} = \frac{2}{\|\mathbf{w}\|}$$

Key move

- Rewrite H_- with zero on the right side before substituting x_+

The two 1s in the numerator have different origins

The numerator is still the line expression evaluated at a point

$$d = \frac{\left| \underbrace{\mathbf{w}^\top \mathbf{x}_+ + b}_{=1 \text{ (point on } H_+)} \quad \underbrace{+1}_{\text{from rewriting } H_-} \right|}{\|\mathbf{w}\|} = \frac{2}{\|\mathbf{w}\|}$$

First 1

- $w^T x + b$ evaluated at the chosen point x_+
- It equals 1 because x_+ lies on H_+

Final +1

- Moving H_-'s right side, −1, to the left
- It belongs to the target line, not the point

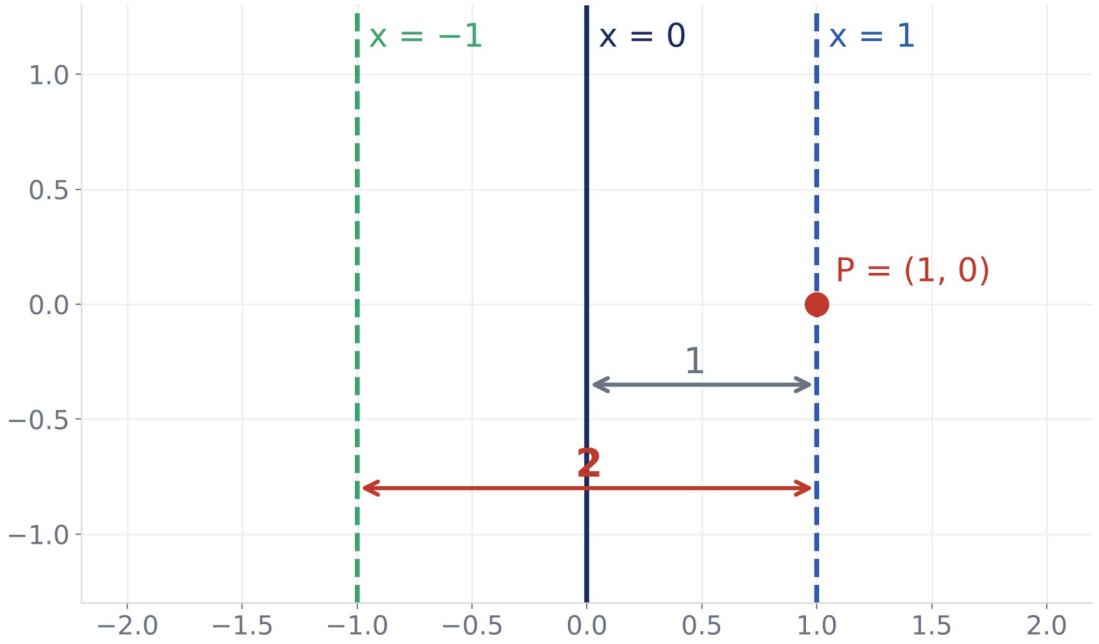
A point on one margin is 2 units from the other

Boundary x = 0, margins x = 1 and x = −1, point P = (1, 0)

$$d = \frac{|1 \cdot 1 + 0 \cdot 0 + 1|}{\sqrt{1^2 + 0^2}} = \frac{2}{1} = 2$$

Steps

- Rewrite x = −1 as x + 1 = 0
- Read a = 1, b = 0, c = 1
- Substitute P = (1, 0)
- Distance to x = 0 is 1



Maximizing 2 / ||w|| is minimizing ½||w||²

Linearly separable data; labels y_i are −1 or +1

$$y_i \left(\mathbf{w}^\top \mathbf{x}_i + b \right) \geq 1, \qquad y_i \in \{-1, +1\}$$

$$\max_{\mathbf{w}, b} \frac{2}{\|\mathbf{w}\|} \iff \min_{\mathbf{w}, b} \frac{1}{2} \|\mathbf{w}\|^2$$

Constraint

- Every point lies on or outside its margin

Why ½||w||²

- Same optimum, smooth and convex

The ±1 scaling gives the norm its geometric meaning

Rescaling w and b by the same positive k leaves the boundary unchanged

$$(\mathbf{w},\, b) \;\rightarrow\; (k\mathbf{w},\, kb),\; k > 0: \qquad k\mathbf{w}^\top \mathbf{x} + kb = 0 \iff \mathbf{w}^\top \mathbf{x} + b = 0$$

The problem

- Any k > 0 gives the same boundary
- Without a fix, ||w|| could be made arbitrarily small

The fix

- Require the closest points to satisfy $y_i(w^T x_i + b) = 1$
- Then ||w|| measures the margin: width = 2 / ||w||

Soft margin allows violations at a price

Data may overlap; ξ_i are slack variables

$$\min_{\mathbf{w}, b, \boldsymbol{\xi}} \frac{1}{2}\|\mathbf{w}\|^2 + C\sum_{i=1}^{n}\xi_i \quad \text{s.t.} \quad y_i(\mathbf{w}^\top \mathbf{x}_i + b) \geq 1 - \xi_i, \quad \xi_i \geq 0$$

Slack ξ_i

- How far point i violates its margin
- $\xi_i = 0$ means no violation

Penalty C

- Large C: strict, close to hard margin
- Small C: wider margin, more violations

Soft margin equals regularization plus hinge loss

Same optimum as the soft-margin form, with ξ_i eliminated

$$\min_{\mathbf{w},\,b}\ \frac{1}{2}\|\mathbf{w}\|^2 + C\sum_{i=1}^{n}\max\bigl(0,\ 1 - y_i(\mathbf{w}^\top\mathbf{x}_i + b)\bigr)$$

Hinge loss

- Zero when $y_i(w^Tx_i + b) \geq 1$
- Grows linearly inside the margin

Why this view helps

- Same template as other loss-based models
- $\frac{1}{2}||w||^2$ acts as the regularizer

The dual uses the data only through inner products

Used to extend SVM to kernels

$$\max_{\boldsymbol{\alpha}} \sum_i \alpha_i - \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j \, y_i y_j \, \mathbf{x}_i^\top \mathbf{x}_j \quad \text{s.t.} \quad 0 \le \alpha_i \le C, \quad \sum_i \alpha_i y_i = 0$$

Kernel trick

- Replace $x_i^T x_j$ with $K(x_i, x_j)$
- This gives a nonlinear SVM

Support vectors

- Points with $\alpha_i > 0$
- They alone determine the boundary

Four forms describe one maximum-margin problem

Each form serves a different purpose

Form	Objective	When to use
Hard margin	Maximize the margin; no violations	Separable data
Soft margin	Margin plus C × total slack	Overlapping data
Hinge loss	Regularizer plus C × hinge loss	Loss-function view
Dual	Inner products weighted by α_i	Kernels and support vectors

The objective follows from one distance rule

Point-to-line distance, ±1 scaling, and the norm of w

1

Margin = 2 / ||w||

The point-to-line distance in vector form, applied between H_+ and H_-.

2

Maximize margin ⇔ minimize ½||w||²

The ±1 scaling fixes the freedom to rescale (w, b).

3

Soft margin, hinge loss, and the dual extend it

Slack handles overlap; the dual brings kernels and support vectors.

One geometric idea, four ways to write the optimization